Image Representation

MSIS 521

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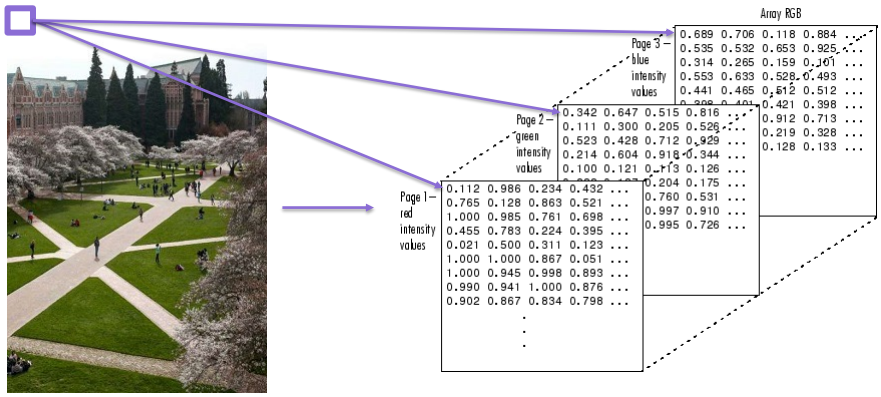
3D Array Representation



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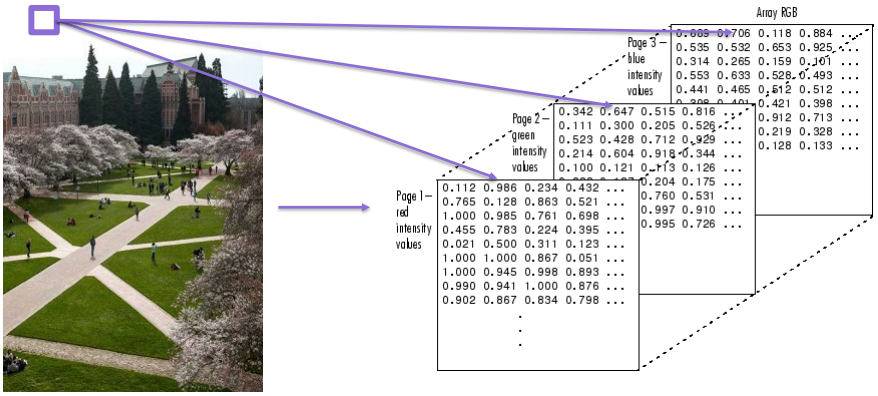
3D Array Representation



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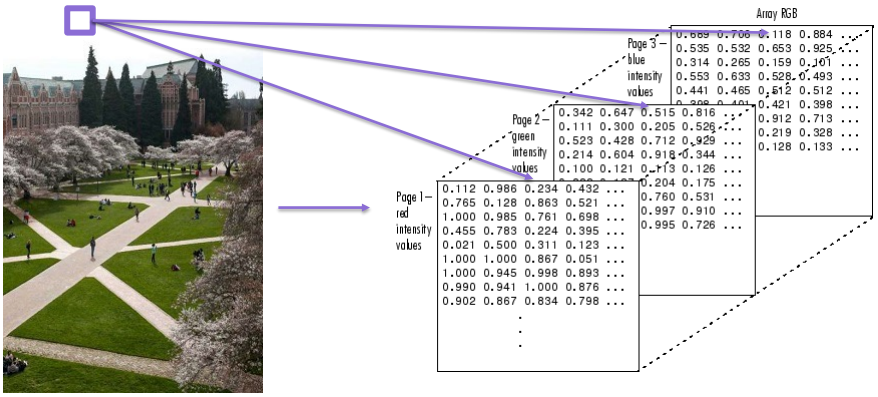
3D Array Representation



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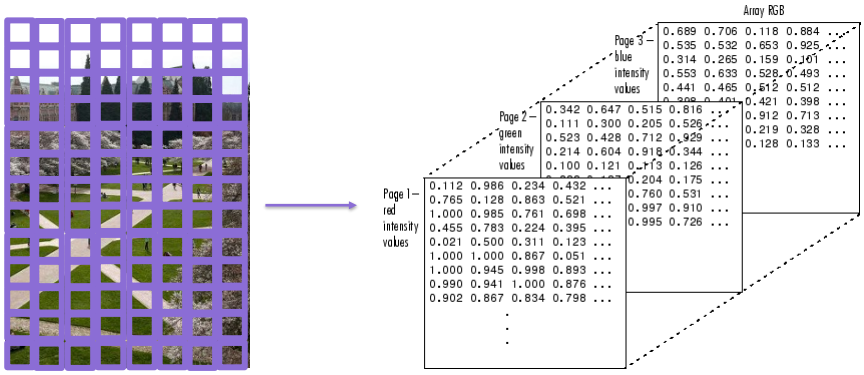
3D Array Representation



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3D Array Representation

- Row by row
- # Dimensions = image height × image width
- Basic unit: each cell = a pixel
- Each pixel corresponds to [R,G,B] three intensity values
- Each value is taken from the interval of (0,255) or (0,1)

- Not the same length of dimensions for different image documents

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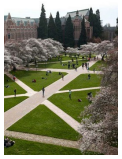
Basic Image Manipulations

- Crop **
- Resize **
- (Rotate)
- Plot



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Practice

```
from PIL import Image
import matplotlib.pyplot import as plt

##image manipulation
im = Image.open('uw.png')

crop = (75,100,180,150)
im_crop = im.crop(crop)
im_rs = im.resize((100,100))
im_rotate = im.rotate(180)

print(im)
print(im_crop)
print(im_rs)
print(im_rotate)
```

```
imshow(im)
imshow(im_crop)
imshow(im_rs)
imshow(im_rotate)
```

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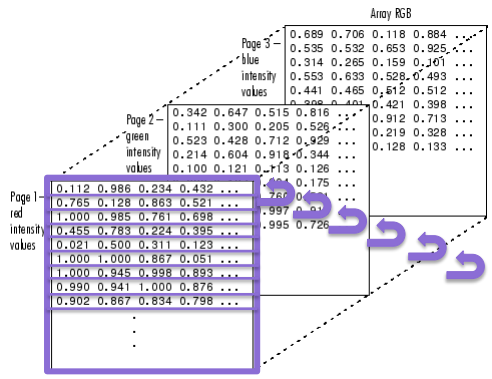
Math Representation

- Array representation: 3D
  - Array: I[:,:,:]
- What is?
  - I[20:40,10:100,:]
  - 255-I



- Keep grayscale only: 2D

- Vectorization: 1D

[0.40392157, 0.39607843, 0.4, ..., 1.,1.,1.]

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Practice

```
##math representation
im_m = array(im)
im_gray_m = array(im.convert('L'))
im_v = im_m.flatten()

imshow(im_gray_m, cmap = plt.cm.gray)


print(im_m)
print(im_gray_m)
print(im_v)

im_m.shape
im_v.shape
im_gray_m.shape
```

```
im_4 = im_m[:,:,3]
imshow(im_4)

im_s = im_m[20:60,:,:]
imshow(im_s)

im_i = 255 - im_m[:,:,:-1]
imshow(im_i)

im_3d = im_m[:,:,:-1]
imshow(im_3d)

figure()
hist(im_v,256)
```

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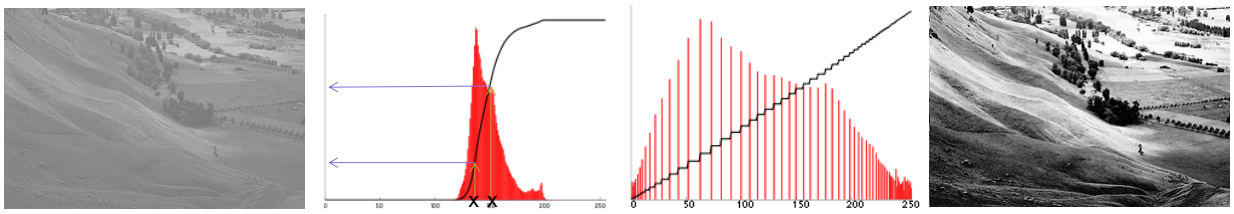
Image Normalization

- Histogram Equalization
  - Flattens the (graylevel) histogram of an image so that all **intensities** are as **equally common** as possible
  - Enhances contrast



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Practice

```
##normalization
imhist, bins = histogram(im_gray_m.flatten(), 256, density=True)
cdf = imhist.cumsum()
cdf = 255 * cdf / cdf[-1]
im2 = interp(im_gray_m.flatten(), bins[:-1], cdf)
im_norm = im2.reshape(im_gray_m.shape)

##display
fig, ax = plt.subplots(1, 2, sharey = True)
ax[0].imshow(im_norm, cmap = plt.cm.gray)
ax[1].imshow(im_gray_m, cmap = plt.cm.gray)
ax[0].set_title('Normalized')
ax[1].set_title('Original')
fig.savefig("norm_pic.png")
```

```
##plot histograms
figure()
hist_0 = hist(im_gray_m.flatten(), 256)
hist_n = hist(im2.flatten(),256)
```

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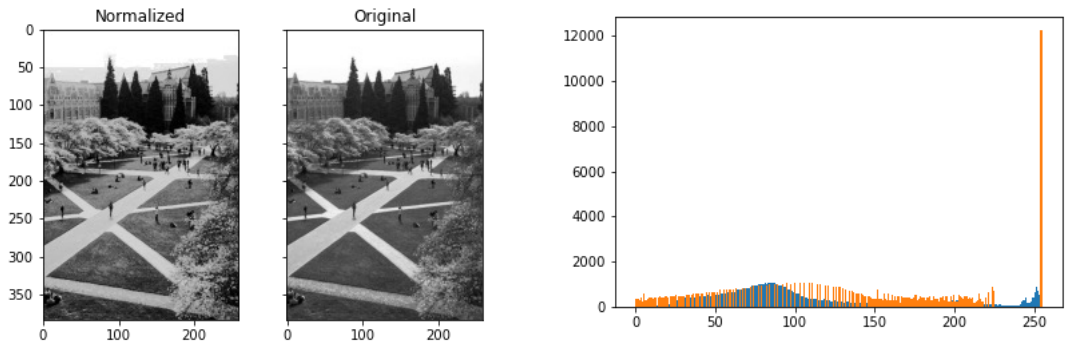
Results



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